بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

Cs_101

Introduction to Computing

Current Mid_Term PAPERs
Fall 2022
Full Solution

ZOOLOGY
LEGENDS

"Don't Stop When you Tired,
Stop When you Done"

0304-4756496

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Mcqs:

- **Pop3 stands for?**

  Post Office Protocol Version 3

- **SGML stands for?**

  Standard Generalized Markup Language,

- **In hexadecimal notation 1011 represent?**

  1011 represent 11 in Binary but here it is representing 3 (excess of 8).

- **which refers to single sender involved with multiple unicasts**

  multicast

- **Usually IP address was a pattern of how many bits?**
  - **32 bits**
- **The simplest of these is the repeater, which is little more than a device that passes signals back and forth between the two original buses**
  - **Repeater**
- **what is the symbol of wild card?**

  “* ”

- **---- gives you advertisement based on your search queries and history?**

  Google

- **which notation uses a single symbol to represent a pattern of four bits?**

  Hexadecimal notation

- **how many manageable unit cell in the main memory?**

  8 bits

- **which devices are designed to take place of magnetic disks?**

  SSDs (Solid State Disks): (Larger Flash memory devices designed to take place of magnetic disk)

- **Which of the following language is used for client side scripting?**

  ____JavaScript

- **which OS software is used to store the files in memory.**

- **ANCI**

  American National Standards Institute (ANSI)

  American Standard Code for Information Interchange (ASCII)

- **An enormous portion of current internet traffic is used for transporting audio and video across the internet in real time, is known as**

Streaming

- **How many bits a byte have**

  1 byte = 8 bits

- **USB stands for ?**

  - **Universal serial bus**

- **What is TCP abbreviation ?**

  Transmission Control Program

- **Symbol which displays to record voice { google voice typing ?**

- **DNS stands for**

  Domain name system

- **GUI stands for**

  Graphical user interface

- **A long string is often called a———**

  ANSII STRING

- **a sequence of transactions in which the user establishes initial contact with a computer's operating system**

  Login procedure

- **USB and FireWire are examples of high-speed data transfer over a .....**

  Short distance

- It is the _____job to decide in which direction a packet should be sent at each step along the packet's path through the Internet.

  Network layer's

- Communication between a computer and other devices is normally handled through an intermediary apparatus known as a_____.

  Controller

- A typical dictionary in a word processor contains approximately ____entries

  25,000

- URL

  Uniform Resource Locator

- Email is less time sensitive, so the mail servers use_____to transfer email

  SMTP

  - SMTP for text messages.
  - MIME (Multipurpose Internet Mail Extensions) to convert non-ASCII to SMTP compatible form.
  - Post Office Protocol Version 3 (POP3)
  - Internet Mail Access Protocol (IMAP)
  - POP3 helps to store messages on local machine,
  - IMAP on mail server machine

- **A clear flag (value 0) indicates that the printer is available and a set flag (value 1) indicates that the printer is currently allocated.**
-

Short Questions:

1) Transport layer present in ISO?

The application layer uses the Transport layer to send and receive messages over the internet. The Transport layer divides long messages into small segments, which are transmitted over the Internet. An important task of the transport layer is to accept messages from the application layer and to ensure that the messages are properly formatted for transmission over the Internet

2) Use two protocols name them. One is realible why?

1- In a bus network based on the Ethernet standards, the right to transmit messages is controlled by the protocol known as **Carrier Sense, Multiple Access with Collision Detection (CSMA/CD).** This protocol dictates that each message be broadcast to all the machines on the bus

2- Another cause might be that the signals from the different machines are blocked from each other by objects or distance even though they can all communicate with the

central AP. The result is that wireless networks adopt the policy of trying to avoid collisions rather than trying to detect them. Such policies are classified as **Carrier Sense, Multiple Access with Collision Avoidance (CSMA/CA),** many of which are standardized by IEEE

3) Binary addition of two numbers, Two values given (Repeated)

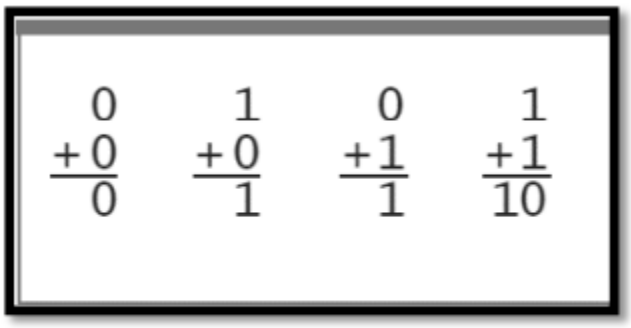


Figure 26: Binary Addition basics

4) Write the basic three types of Arithmetic Logic Unit.

1- Arithmetic/logic unit,

2- Control unit

3- Register unit

5) What is windows manager? Explain its types.

Window manager, which allocates blocks of space on the screen, called windows, and keeps track of which application is associated with each window

TyPes:

- File manager
- Device drivers

- Memory manager

6) Packets

The transport layer adds sequence numbers to the small segments it produces so that the segments can be reassembled at the message's destination. Then it hands these segments, known as packets, to the network layer.

7) Name three conditions if they are satisfied then won't result in deadlock

Deadlock has revealed that it cannot occur unless all three of the following conditions are satisfied:

1. There is competition for non-sharable resources.

2. The resources are requested on a partial basis; that is, having received some resources, a process will return later to request more.

3. Once a resource has been allocated, it cannot be forcibly retrieved.

8) 1-List Search Engines except google.

Google, Yahoo, Firefox, Chrome, Microsoft Edge and MSN

9) Print server & file server

Print server: and the other machines were programmed to play the role of clients that sent print requests to the print server.

File server: The other machines played the role of clients that requested access to the files that were stored at the file server.

10) Queries google:

Pslcricket@Twitter: particular social media, you can add "@"

#Artificial intelligence: On Social media, sometimes, a hash tag is very popular, if you are interested to identify such pages using hash tag, you can use the query like: "#education"

***for tat:** Wild Card based Searching "* is thicker than water"

11) Lossy types:

Lossy schemes are those that may lead to the loss of information. Lossy techniques often provide more compression than lossless ones and are therefore popular in settings in which minor errors can be tolerated, as in the case of images and audio. In cases where the data being compressed consist of long sequences of the same value, the compression technique called **run-length encoding**, which is a lossless method, is popular.

Another lossless data compression technique is **frequency-dependent encoding**, a system in which the length of the bit pattern used to represent a data item is inversely related to the frequency of the item's use.

12) Tier-1 ISO.

At the top of this hierarchy are relatively few tier-1 ISPs that consist of very high- speed, high-capacity, international WANs. These networks are thought of as the backbone of the Internet. They are typically operated by large companies that are in the communications business. An example would be a company that originated as a traditional telephone company and has expanded its scope into providing other communication services.

13) Pheriphal devices

Mice, printers, scanners, mass storage devices, digital cameras, and smartphones.

14) Sharing time

With the development of multiuser, time-sharing operating systems, a typical computer installation was configured as a large central computer connected to numerous workstations. From these workstations, users could communicate directly with the computer from outside the computer room rather than submit- ting requests to a computer operator

15) Internet Multimedia Streaming

Internet Multimedia Streaming An enormous portion of current Internet traffic is used for transporting audio and video across the Internet in real-time, known as streaming. Netflix streamed more than 4 billion hours of programming to end users in the first three months of 2013 alone.

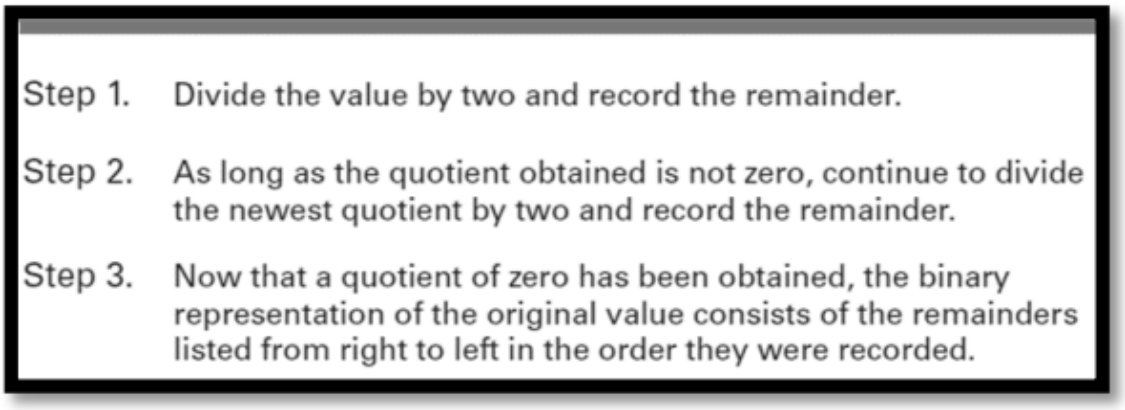
16) Names three types of image compression

- GIF (short for Graphic Interchange Format)
- JPEG (Joint Photographic Experts Group)
- TIFF (short for Tagged Image File Format)

17) Machine instruction

This collection of instructions along with the encoding system is called the machine language. An instruction expressed in this language is called a machine level instruction or, more commonly, a machine instruction

18) Algorithm for converting positive decimal to binary number



Step 1. Divide the value by two and record the remainder.

Step 2. As long as the quotient obtained is not zero, continue to divide the newest quotient by two and record the remainder.

Step 3. Now that a quotient of zero has been obtained, the binary representation of the original value consists of the remainders listed from right to left in the order they were recorded.

Long Questions:

- **Url stands for what? Give three examples.**

Uniform Resource Locator (URL)

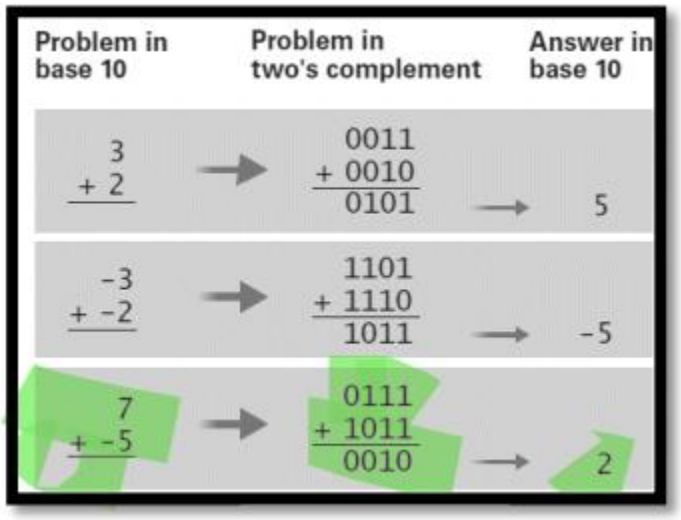
1- INurl
2- Allurl

- **Convert hexadecimal value to binary bit notation.**

Table 7: Hexadecimal Notation

Bit pattern	Hexadecimal representation
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

Bit pattern	Value represented
1111	7
1110	6
1101	5
1100	4
1011	3
1010	2
1001	1
1000	0
0111	-1
0110	-2
0101	-3
0100	-4
0011	-5
0010	-6
0001	-7
0000	-8



- **What is meant by CDNs. Explain it's main purpose?**

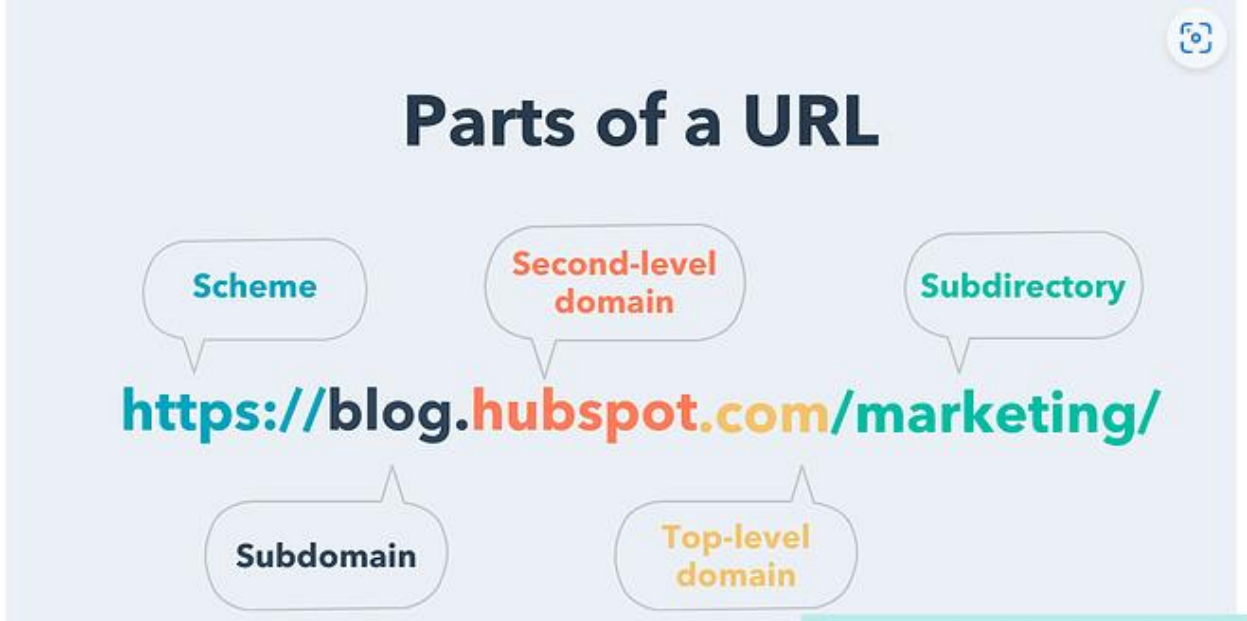
A content delivery network (CDN) is a **distributed network of servers that can efficiently deliver web content to users**. CDNs store cached content on edge servers in point-of-presence (POP) locations that are close to end users, to minimize latency.

- **8 bit binary numbers to floating point notation.**

Today use at least 32 bits for storing values in floating-point notation instead of the 8 bits we have used here.

00001000

- **Write two basic segments of URL.**



- **"Good luck" convert into ASCII**

G = 01000111

O = 01101111

o = 01101111

d = 01100100

= 00100000

L = 01001100

u = 01110101

c = 01100011

k = 01101011

ASCII - Binary Character Table

Letter	ASCII Code	Binary	Letter	ASCII Code	Binary

a	097	01100001	A	065	01000001
b	098	01100010	B	066	01000010
c	099	01100011	C	067	01000011
d	100	01100100	D	068	01000100
e	101	01100101	E	069	01000101
f	102	01100110	F	070	01000110
g	103	01100111	G	071	01000111
h	104	01101000	H	072	01001000
i	105	01101001	I	073	01001001
j	106	01101010	J	074	01001010
k	107	01101011	K	075	01001011
l	108	01101100	L	076	01001100
m	109	01101101	M	077	01001101
n	110	01101110	N	078	01001110
o	111	01101111	O	079	01001111
p	112	01110000	P	080	01010000
q	113	01110001	Q	081	01010001
r	114	01110010	R	082	01010010
s	115	01110011	S	083	01010011
t	116	01110100	T	084	01010100
u	117	01110101	U	085	01010101
v	118	01110110	V	086	01010110
w	119	01110111	W	087	01010111

x	120	01111000	X	088	01011000
y	121	01111001	Y	089	01011001
z	122	01111010	Z	090	01011010

- **Table tha jis mn bits ka decimal number likhna tha.**

Table 11: Decimal System representation

Hundreds	Tens	Units
$10^2 = 100$	$10^1 = 10$	$10^0 = 1$
3	7	5

- **Query weather sibi, currency US, divide Question 1000 divided by 30.**

"Weather **sibi**"

For example, you want to see how much 100 Dollars will be when converted to Pakistani rupees.

1000/30

- **Cloud computing long.**

Cloud computing, whereby huge pools of shared computers on the network can be allocated for use by clients as needed, is the latest trend in distributed systems.

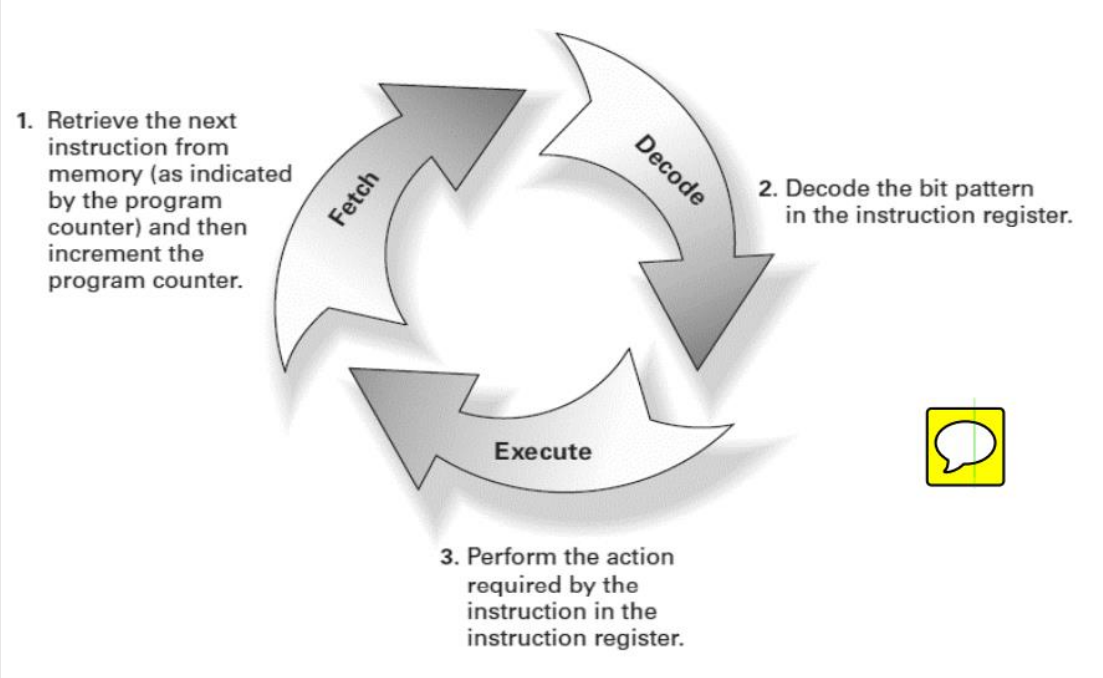
Cloud computing services provide reasonable guarantees of reliability and scalability, but also raise concerns about privacy and security in a world where we may no longer know who owns and operates the computers that we use.

Google Drive and Google Apps

- **JPEG**

Another popular compression system for images is JPEG (pronounced "JAY-peg"). It is a standard developed by the Joint Photographic Experts Group (hence the standard's name) within ISO. JPEG has proved to be an effective stan- dard for compressing color photographs and is widely used in the photography industry, as witnessed by the fact that most digital cameras use JPEG as their default

- **write any name of machine cycle describe of details of cycle**



- **There are several working ISP in Pakistan. write any name of working ISP in Pakistan**

The Internet is a collection of connected net- works. In general, these networks are constructed and maintained by organizations called Internet Service Providers (ISPs)

1- At the top of this hierarchy are relatively few tier-1 ISPs that consist of very high- speed, high-capacity, international WANs. These networks are thought of as the backbone of the Internet.
Example: Traditional Telephone Company

2- Connecting to the tier-1 ISPs are the tier-2 ISPs that tend to be more regional in scope and less potent in their capabilities. (The distinction between the tier-1 and tier-2 ISPs is often a matter of opinion.)

3- Tier-1 and tier-2 ISPs are essentially networks of routers that collectively provide the Internet's communication infrastructure. As such, they can be thought of as the core of the Internet. Access to this core is usually provided by an intermediary called an access or tier-3 ISP
Examples include cable and telephone companies, telephones, video cameras, automobiles, and home appliances.

- **Machine's instructions can be categorized into three groupings:**

1. **Data Transfer Group** The data transfer group consists of instructions that request the movement of data from one location to another. Instructions handle the input/output (I/O) activities of the machine; they are called I/O instructions and are sometimes considered as a category in their own right.

2. **Arithmetic/Logic Group** The arithmetic/logic group consists of the instructions that tell the control unit to request an activity within the arithmetic/logic unit.
3. **Control Group** The control group consists of those instructions that direct the execution of the program rather than the manipulation of data.

There is all are 100% right solved

But there is a little chance of any mistake

Please If You find any mistake then also let us to Know

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JazakAllaH